

Exercise 22

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Introduction to Electrodynamics

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Step 1

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We know that the electric field of an infinite line of charge, charge density λ , is:

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{s}$$

Thus the potential is:

$$V(s) = - \int_{s_0}^s \frac{\lambda}{2\pi\epsilon_0 s} ds = \boxed{-\frac{\lambda}{2\pi\epsilon_0} \ln \frac{s}{s_0}}$$

Now take the gradient and check the result:

$$-\nabla V = \frac{\lambda}{2\pi\epsilon_0} \hat{s} \frac{\partial}{\partial s} (\ln s - \ln s_0) = \frac{\lambda}{2\pi\epsilon_0 s} \hat{s} = \vec{E}$$

, confirming our solution.

Result

2 of 2

$$\boxed{V(s) = -\frac{\lambda}{2\pi\epsilon_0} \ln \frac{s}{s_0}}$$

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